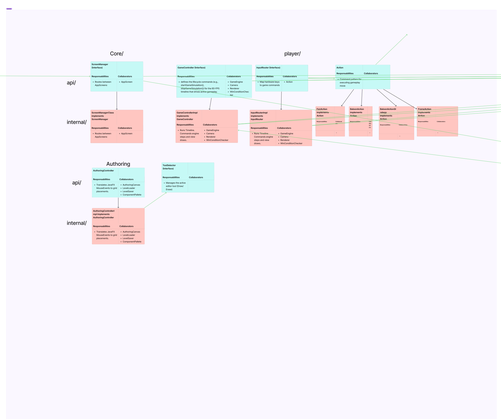
Core/
api/
internal/
Core/Engine/
Engine.h
Responsibilities
Initialize subsystems
Run the main loop
Shut down cleanly
Collaborators
Application
Core/Interfaces/
IGameSystem.h
Responsibilities
Define the lifecycle contract for all systems
Collaborators
None (pure interface)
Core/Events/
EventBus.h
Responsibilities
Decouple publishers from subscribers
Dispatch events to registered handlers
Collaborators
Any system that publishes or subscribes
Core/Events/
EventTypes.h
Responsibilities
Define all event structs
Collaborators
None
Core/Config/
ConfigManager.h
Responsibilities
Load and parse config files
Provide typed access to settings
Collaborators
None
Core/Logging/
Logger.h
Responsibilities
Write log messages to output
Support log levels
Collaborators
None
player/
api/
internal/
Player/
PlayerController.h
Responsibilities
Translate input into player actions
Collaborators
InputState
Player/
PlayerState.h
Responsibilities
Track health, position, and inventory
Collaborators
None
Player/
PlayerSystem.h
Responsibilities
Update player each frame
Apply damage and healing
Collaborators
PlayerState
Core/Events/EventBus
Player/
PlayerAnimator.h
Responsibilities
Select animation clips based on player state
Collaborators
PlayerState
Animation/AnimationClip
Player/
PlayerCamera.h
Responsibilities
Follow the player with smooth lag
Collaborators
PlayerState
Core/Math/Vec2
Authoring
api/
internal/
Authoring/Commands/
ICommand.h
Responsibilities
Define execute and undo
Collaborators
None
Authoring/Commands/
CommandStack.h
Responsibilities
Maintain undo and redo stacks
Collaborators
ICommand
Authoring/Commands/
PlaceTileCommand.h
Responsibilities
Place a tile and remember what was there
Collaborators
ICommand
World/TileMap
Authoring/Commands/
PlaceEntityCommand.h
Responsibilities
Spawn an entity and remember its data
Collaborators
ICommand
World/EntityManager
Authoring/Commands/
DeleteEntityCommand.h
Responsibilities
Remove an entity and remember its data
Collaborators
ICommand
World/EntityManager
Authoring/Commands/
MoveEntityCommand.h
Responsibilities
Move an entity and remember its old position
Collaborators
ICommand
World/EntityManager
Authoring/Commands/
ResizeMapCommand.h
Responsibilities
Change map dimensions and remember old tiles
Collaborators
ICommand
World/TileMap
Authoring/Commands/
PaintRegionCommand.h
Responsibilities
Paint a rectangular region and remember old tiles
Collaborators
ICommand
World/TileMap
Authoring/Commands/
CompositeCommand.h
Responsibilities
Group multiple commands into one undo step
Collaborators
ICommand
Authoring/Commands/
CommandStack.h
Authoring/Tools/
ToolManager.h
Responsibilities
Manage the active tool
Route input to the active tool
Collaborators
ITool
Authoring/Tools/
ITool.h
Responsibilities
Define the tool interface
Collaborators
None
Authoring/Tools/
TilePaintTool.h
Responsibilities
Paint tiles while the mouse is held down
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileEraseTool.h
Responsibilities
Erase tiles while the mouse is held down
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
EntityPlaceTool.h
Responsibilities
Place an entity at the cursor
Collaborators
ITool
Authoring/Commands/PlaceEntityCommand
Authoring/Tools/
EntitySelectTool.h
Responsibilities
Select and move entities
Collaborators
ITool
Authoring/Commands/MoveEntityCommand
Authoring/Tools/
RectSelectTool.h
Responsibilities
Select entities in a rectangular area
Collaborators
ITool
Authoring/Tools/
TileFillTool.h
Responsibilities
Flood fill a region with a tile
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileSampleTool.h
Responsibilities
Pick a tile from the map
Collaborators
ITool
Authoring/Tools/
TileSelectTool.h
Responsibilities
Select a rectangular region of tiles
Collaborators
ITool
Authoring/Tools/
TileStampTool.h
Responsibilities
Stamp a multi-tile pattern
Collaborators
ITool
Authoring/Tools/
TileReplaceTool.h
Responsibilities
Replace all tiles of one type with another
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileLineTool.h
Responsibilities
Draw a line of tiles
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileRectTool.h
Responsibilities
Draw a rectangle of tiles
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileCircleTool.h
Responsibilities
Draw a circle of tiles
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileBucketTool.h
Responsibilities
Fill a region with a tile
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileEyedropperTool.h
Responsibilities
Sample a tile and set it as the active brush
Collaborators
ITool
Authoring/Tools/
TileSampleTool
Authoring/Tools/
TileRandomTool.h
Responsibilities
Paint tiles randomly from a weighted set
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileMirrorTool.h
Responsibilities
Mirror tiles horizontally or vertically
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileRotateTool.h
Responsibilities
Rotate a selection of tiles
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileSelectTool
Authoring/Tools/
TileCopyTool.h
Responsibilities
Copy a selection to a clipboard
Collaborators
ITool
Authoring/Tools/
TileSelectTool
Authoring/Tools/
TilePasteTool.h
Responsibilities
Paste clipboard contents at cursor
Collaborators
ITool
Authoring/Tools/
TileCopyTool
Authoring/Tools/
TileCutTool.h
Responsibilities
Cut a selection to clipboard
Collaborators
ITool
Authoring/Tools/
TileSelectTool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileFlipTool.h
Responsibilities
Flip a selection horizontally or vertically
Collaborators
ITool
Authoring/Tools/
TileSelectTool
Authoring/Tools/
TileScatterTool.h
Responsibilities
Scatter tiles randomly within a region
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileRandomTool
Authoring/Tools/
TileOutlineTool.h
Responsibilities
Draw the outline of a region
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileRectTool
Authoring/Tools/
TileGradientTool.h
Responsibilities
Fill a region with a gradient of tiles
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileFillTool
Authoring/Tools/
TileNoiseTool.h
Responsibilities
Paint tiles using noise patterns
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileRandomTool
Authoring/Tools/
TileSymmetryTool.h
Responsibilities
Paint with symmetry enabled
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileMirrorTool
Authoring/Tools/
TileStencilTool.h
Responsibilities
Paint through a stencil mask
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileSelectTool
Authoring/Tools/
TileMeasureTool.h
Responsibilities
Measure distances and areas
Collaborators
ITool
Authoring/Tools/
TileSelectTool
Authoring/Tools/
TileAlignTool.h
Responsibilities
Align a selection to edges or grid
Collaborators
ITool
Authoring/Tools/
TileSelectTool
Authoring/Tools/
TileJitterTool.h
Responsibilities
Add random offset to painted tiles
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileRandomTool
Authoring/Tools/
TileSmudgeTool.h
Responsibilities
Smudge tiles like a finger tool
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileSampleTool
Authoring/Tools/
TileBlendTool.h
Responsibilities
Blend tile edges with neighbors
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileAutoTool
Authoring/Tools/
TileAutoTool.h
Responsibilities
Auto-tile based on neighbor rules
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileRandomTool
Authoring/Tools/
TileSmoothTool.h
Responsibilities
Smooth tile edges
Collaborators
ITool
Authoring/Commands/PaintRegionCommand
Authoring/Tools/
TileNoiseTool
Authoring/Tools/
TileWeightedTool.h
Responsibilities
Paint tiles with weighted probability